

ChefGPT

Ai Powered Recipe Generator

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4th Internship / Project Report

13 September 2025

Problem Statement

- Cooking can be time-consuming, and finding recipes based on available ingredients is challenging.
- Existing apps may not generate creative recipes dynamically.
- Goal: Create an AI-powered recipe app that generates recipes based on ingredients or free prompts and provides images.

Project Overview

ChefGPT Features:

- User authentication: Register, login, logout (JWT + cookies)
- Two input modes:
 1. Ingredients-based
 2. Name/description or free text
- Generates 3 dish suggestions with:
 1. Name
 2. Description
 3. AI-generated image prompt
- Fetch full recipe + images on selection
- Add/Remove favorite dishes
- Favorites tab for all saved dishes

Architecture Overview

- Frontend: React , Tailwind, Shadcn UI
- Backend: FastAPI
- Database: PostgreSQL (SQLAlchemy ORM)
- AI/Image APIs:
 - OpenAi (recipe generation)
 - StableHorde (image generation)
 - Cloudinary (image upload & storage)
- Authentication: JWT + secure cookies

Tech Stack

- Backend: Python, FastAPI, SQLAlchemy, JWT
- Database: PostgreSQL
- Recipe/Image Generation : OpenAi, StableHorde API,
- Image Storage: Cloudinary
- Environment Management: .env files for secrets
- Testing & Dev: Pydantic, async Python, Postman

User Authentication

- Features implemented:
 - Register → store username, email, hashed passwords (bcrypt)
 - Login → generate JWT token
 - Logout → clear cookie
- Security measures:
 - Secure cookies
 - JWT expiry
 - Input validation with Pydantic schemas

Recipe Generation Workflow

- Mode 1: Ingredients-based
 - User inputs available ingredients → AI generates 3 recipe suggestions
- Mode 2: Name/Description
 - User inputs dish name/description → AI generates 3 recipe suggestions
- Steps:
 - Input → API → OpenAi → return recipe + image prompt
 - Prompt → API → StableHorde → return Image
 - Upload image to Cloudinary
 - Display results to user

Favorite Flow

- Add/Remove favorites
 - User can save AI-generated dishes
 - Favorite dishes stored in PostgreSQL
- Favorites tab
 - Displays all saved dishes
 - Supports removal
- Flow Diagram: Dish → Add to Favorites → DB → Favorite Tab

Key challenges

- Handling different image input formats (URL, base64, Cloudinary)
- Managing JWT auth with cookies securely
- Integrating AI API (StableHorde, OpenAi) and Cloudinary together
- Ensuring database consistency for favorites
- Async handling in FastAPI + image uploads

Key challenges

- ChefGPT is a full-stack AI-powered recipe generator.
- Handles auth, AI integration, image hosting, and favorites management.
- Demonstrates skills in:
 - FastAPI, async Python
 - Cloud services (Cloudinary)
 - API integration (StableHorde, OpenAi)
 - JWT auth + secure cookies
- Ready for deployment and extension.

Thank You!